

Matrix Spencer, twice: a comparison of arXiv:2608.28816 and arXiv:2609.15025

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Abstract

This note compares two recent arXiv papers on the Matrix Spencer conjecture: *A Proof of the Matrix Spencer Conjecture* by Emrullah Akbas and the author (arXiv:2608.28816, posted 28 August 2026), and *Matrix Spencer: Eight Standard Deviations Suffice and an Almost-Linear Time Algorithm for Dense Input* by Zhao Song and Lichen Zhang (arXiv:2609.15025, posted 14 September 2026). It records the chronology, places what the second paper says about the first next to the public record, and compares the two proofs section by section. The comparison shows the same barrier family, the same diagonal-to-full interpolation of a log-partition function, and the same matrix-weighted Poincaré pipeline. It also identifies what is genuinely new in the second paper. The note closes with the corrections the record requires. Every quotation and every section, definition, and lemma number below is taken from the posted v1 of each paper and can be checked against it.

1 Chronology and claims

The public record consists of two arXiv postings.

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| 28 Aug 2026 | [AS26] v1. Theorem 1.2 (<i>Matrix Spencer</i>): for symmetric $n \times n$ matrices $A_1, \dots, A_n$ with $\ A_i\ _{\text{op}} \leq 1$, $\min_{x \in \{\pm 1\}^n} \ \sum_i x_i A_i\ _{\text{op}} \leq C\sqrt{n}$; “moreover, a coloring with discrepancy at most $C\sqrt{n}$ can be found efficiently.” Theorem 1.3 (<i>Hereditary Matrix Small Ball</i>) is the engine; the abstract singles it out as “a hereditary small-ball estimate for Gaussian series that should be of independent interest.” |
| 14 Sep 2026 | [SZ26] v1 (title page dated 1 September 2026). Existence of a signing with discrepancy below $8\sqrt{n}$; a randomized $n^{3+o(1)}$ polylog($1/p$)-time algorithm with discrepancy below $12\sqrt{n}$. |
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What [SZ26] says about the state of the problem. The abstract: “the conjecture was known only under rank, block-diagonal, or Frobenius-norm restrictions. We prove it.” The introduction, after listing the restricted results of Hopkins–Raghavendra–Shetty, Dadush–Jiang–Reis, and Bansal–Jiang–Meka: “The general case has remained open.” The abstract again: “The existence proof is a partial-coloring argument with one new estimate: a hereditary Gaussian small-ball bound.” Section 2.1 repeats it: proving the conjecture “needs one new estimate inside a known frame.”

What [SZ26] says about [AS26]. The paper cites [AS26] exactly twice. Once in the bibliography, and once in the section “Acknowledgments & AI Disclosure” on page 16:

“Independent and Concurrent Work. After completing a draft of this paper, the authors became aware of the independent and concurrent work of [AS26], which proves a similar result. The proof strategy of this paper is fundamentally different from [AS26].”

[AS26] does not appear in the abstract, in the introduction, or in Section 1.2 (“Previous work”), which discusses the restricted results at length. The same page discloses that the authors “have started to prompt AI to get a proof for matrix Spencer” on 28 July 2026, “obtained the existence proof on August 20, 2026,” and that “the proof of the existence of such a signing comes mainly from AI.”

Two statements that are false as posted. On 14 September 2026 the general case had not “remained open” for seventeen days, and the conjecture was not “known only under” restrictions. [SZ26] cites [AS26], so both sentences were posted with knowledge of the result that contradicts them. The rest of this note concerns the third statement, that the proof strategy is “fundamentally different,” and the fourth, that the hereditary small-ball estimate is “new.”

2 Side-by-side comparison of the proofs

Both proofs have the same shape: a partial-coloring frame (Gluskin’s method in [AS26], Rothvoss’s Gaussian projection in [SZ26]) consuming a hereditary small-ball estimate for the Gaussian series $\sum_i g_i A_i$ at radius of order $\sqrt{n}$ times a logarithmic aspect-ratio factor. The frame is classical. The estimate is the content. The relevant comparison is therefore [AS26] Section 3 against [SZ26] Section 4, and it goes as follows.

2.1 The barrier. [AS26] Section 3.1 defines, for an integer q and $\alpha \geq \beta > 0$,

$$v_{q,\alpha,\beta}(t) = \beta \ell(t) + (\alpha - \beta) r_q(t), \quad \ell(t) = -\log(1 - t^2), \quad r_q(t) = \sum_{k \geq q} \frac{t^{2k}}{k}, \quad (1)$$

and the spectral potential $V(B) = \text{Tr } v_{q,\alpha,\beta}(B)$ for $\|B\|_{\text{op}} < 1$, $V(B) = +\infty$ otherwise (their (3.4)). [SZ26] Definition 4.2 (“Barrier potential”) defines, for an integer $\tau \geq 2$ and $\kappa_1 \geq \kappa_0 > 0$,

$$\phi(s) = \kappa_0 \phi_0(s) + (\kappa_1 - \kappa_0) \phi_{1,\tau}(s), \quad \phi_0(s) = -\log(1 - s^2), \quad \phi_{1,\tau}(s) = \sum_{k \geq \tau} \frac{s^{2k}}{k}, \quad (2)$$

extended by $+\infty$ outside $(-1, 1)$, with $\Phi(X) = \text{tr } \phi(X)$. Under

$$q = \tau, \quad \alpha = \kappa_1, \quad \beta = \kappa_0,$$

(1) and (2) are the same function, the constraint $\alpha \geq \beta$ is the constraint $\kappa_1 \geq \kappa_0$, and V is Φ . This is an identity of parameterized functions, not a resemblance between two logarithmic barriers. The design rationale is also the same: weight the low-degree part of $-\log(1 - t^2)$ lightly and the high-degree tail heavily, so that curvature concentrates at the spectral boundary while the bulk is barely penalized ([AS26] Section 3.1, second paragraph; [SZ26] Section 2.1, “The hereditary small-ball estimate”).

2.2 The diagonal-to-full interpolation. [AS26] equations (3.2)–(3.3): fix an orthonormal basis, split $B(g) = B_d(g) + B_o(g)$ into diagonal and off-diagonal parts, and set

$$B_t(g) = B_d(g) + t B_o(g), \quad Z_t = \mathbb{E}_{\gamma_n} [e^{-V(B_t(g))}], \quad F(t) = \log Z_t, \quad 0 \leq t \leq 1.$$

[SZ26] Definition 4.4 (“Diagonal-to-full interpolation”): fix an orthonormal basis, write $A = A_{\text{diag}} + A_{\text{off}}$, and set

$$A_t = A_{\text{diag}} + tA_{\text{off}}, \quad a_t = \mathbb{E}[e^{-\Phi(A_t)}], \quad a(t) = \log a_t, \quad t \in [0, 1].$$

Same object, same role: bound the diagonal endpoint directly, then control the second derivative of the log-partition function along the path.

2.3 The analytic pipeline. The remaining steps correspond one to one.

Mechanism	[AS26]	[SZ26]
Matrix-weighted Gaussian Poincaré inequality	Section 2.1	Section 4.4, Lemma 4.10 (“Matrix-weighted Poincaré criterion”)
Spectral boundary/bulk split of the Hessian; curvature lower bound	Section 3.2, Proposition 3.7 (“Curvature lower bound”)	Section 4.3 (“Spectral Hessian and the 0–1 decomposition”)
Resolvent-weighted Poincaré estimates, iterated over matrix-valued polynomials	Proposition 3.8 and the iteration in Section 3.2 (Lemmas 3.10–3.11)	Sections 4.5–4.7, Lemmas 4.12–4.14
Hereditary small-ball statement	Theorem 1.3	Lemma 4.1 (“Six-regime hereditary small ball”)
Partial-coloring frame	Section 3.5 (Gluskin)	Section 5 (Rothvoss projection)

The vocabulary coincides as well: “hereditary,” “matrix-weighted Poincaré,” “log-partition function,” “interpolation from a diagonal model.”

2.4 What is new in [SZ26]. Three things, and they deserve to be stated plainly.

- (i) *Constants.* [SZ26] carries explicit, certified coefficients through the whole argument (Section 4.8, Appendix A) and reaches discrepancy below $7.8795\sqrt{n}$. [AS26] proves $O(\sqrt{n})$ with an unspecified universal constant.
- (ii) *The hereditary radius.* [AS26] Theorem 1.3 has radius $\kappa\sqrt{n}(1 + \log \frac{2d}{n})^2$; [SZ26] Lemma 4.1 has radius $c_{\text{rad}}\sqrt{m}\ell_{\text{asp}}$ with $\ell_{\text{asp}} = \max\{1, \log(N/m)\}$. Linear rather than squared dependence on the log-aspect-ratio is a genuine sharpening of the estimate.
- (iii) *Running time.* The $n^{3+o(1)}$ algorithm and the sequence of algorithms leading to it (Sections 6–10) have no counterpart in [AS26], which gives a polynomial-time construction.

Conversely, [AS26] Section 2 (a rank-one pinching argument and a dimension-free bound on ratios of partition functions, Theorem 1.4) has no counterpart in [SZ26]. I have not verified the constants, certificates, or algorithms of [SZ26] and make no claim about them here. Improving a constant and a running time are contributions. They are contributions to a theorem that had been proved.

3 Assessment

On “fundamentally different.” Sections 2.1–2.3 show the same barrier family, the same interpolation, and the same Poincaré pipeline, under a change of notation. A proof strategy is not made fundamentally different by sharper constants inside it. If the authors believe there is a fundamental difference, the paper should say what it is. It does not; the sentence on page 16 is the only comparison the paper offers.

On “one new estimate.” The estimate advertised as new in the abstract of [SZ26] is, up to the quantitative improvements in Section 2.4, Theorem 1.3 of [AS26], titled “Hereditary Matrix Small Ball” and highlighted in the abstract of [AS26] seventeen days earlier.

On independence. [SZ26] reports an existence proof obtained on 20 August 2026. I have no basis to dispute this and I do not allege copying. Independent discovery is a fact about how a result was found. It is not a license to tell readers that the problem was open, that the key estimate is new, and that an earlier proof is merely “similar” and “fundamentally different,” in a paragraph most readers will not reach. For the record, [AS26] carries its own “Statement on AI Usage,” crediting GPT-5.6 with the heavy lifting on the hereditary small-ball lemma, and our AI-assisted attack on Matrix Spencer dates to 2025. Which tools a proof came from is not the issue. What the signed account of prior work says is.

4 What the record should say

A corrected v2 of [SZ26] should contain the following.

- (a) Abstract and introduction: a statement that the conjecture was resolved in [AS26] (arXiv:2608.28816, 28 August 2026), in place of “remained open” and “known only under ... restrictions,” with the contribution described as improving the constant and the running time.
- (b) Abstract: attribution of the hereditary Gaussian small-ball estimate to [AS26], with “one new estimate” removed or qualified to the quantitative form of Lemma 4.1.
- (c) Section 1.2: a substantive comparison of Section 4 of [SZ26] with Section 3 of [AS26], listing the shared barrier, interpolation, and Poincaré machinery and stating specifically what differs.
- (d) Acknowledgments: deletion of “fundamentally different,” or its replacement by a specific technical claim.
- (e) Bibliography: a complete entry for [AS26] with arXiv identifier and date.

References

[AS26] E. Akbas and S. Sra. *A Proof of the Matrix Spencer Conjecture*. arXiv:2608.28816, v1 posted 28 August 2026. <https://arxiv.org/abs/2608.28816>

[SZ26] Z. Song and L. Zhang. *Matrix Spencer: Eight Standard Deviations Suffice and an Almost-Linear Time Algorithm for Dense Input*. arXiv:2609.15025, v1 posted 14 September 2026. <https://arxiv.org/abs/2609.15025>